Mathematical models for decision making

Structure of mathematical models

Mathematical models have been developed and used in many application domains, ranging from physics to architecture, from engineering to economics. a model is a selective abstraction of a real system.

In other words, a model is designed to analyze and understand from an abstract point of view the operating behavior of a real system, regarding which it only includes those elements deemed relevant for the purpose of the investigation carried out.

According to their characteristics, models can be divided into *iconic*, *analogical* and *symbolic*.

- **Iconic.** An *iconic* model is a material representation of a real system, whose behavior is imitated for the purpose of the analysis. A miniaturized model of a new city neighborhood is an example of iconic model.
- **Analogical.** An *analogical* model is also a material representation, although it imitates the real behavior by analogy rather than by replication.
- **Symbolic.** A *symbolic* model, such as a mathematical model, is an abstract representation of a real system. It is intended to describe the behavior of the system through a series of symbolic variables, numerical parameters and mathematical relationships.

A further relevant distinction concerns the probabilistic nature of models, which can be either *stochastic* or *deterministic*.

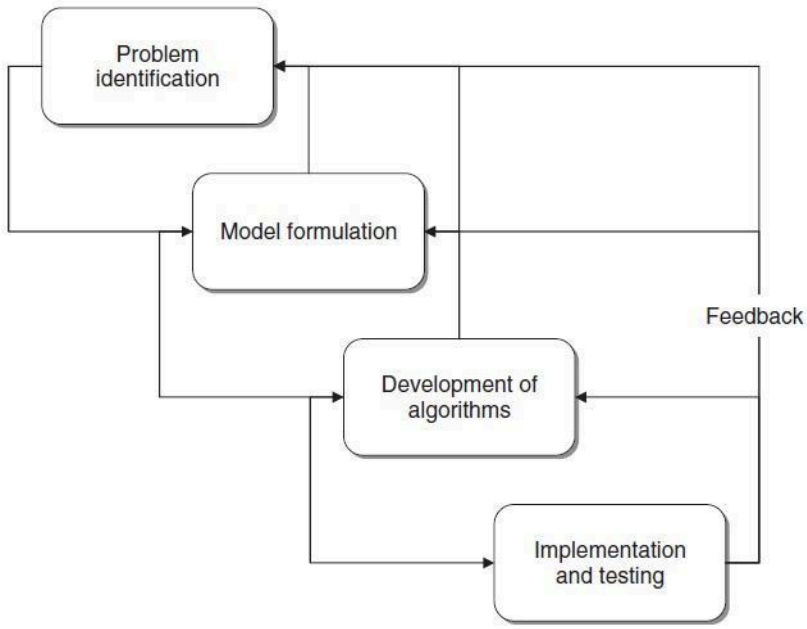
- **Stochastic.** In a *stochastic* model some input information represents random events and is therefore characterized by a probability distribution, which in turn can be assigned or unknown.
- **Deterministic.** A model is called *deterministic* when all input data are supposed to be known a priori and with certainty. Since this assumption is rarely fulfilled in real systems, one resorts to deterministic models when the problem at hand is sufficiently complex and any stochastic elements are of limited relevance.

A further distinction concerns the temporal dimension in a mathematical model, which can be either *static* or *dynamic*.

- **Static.** *Static* models consider a given system and the related decision-making process within one single temporal stage.
- **Dynamic.** *Dynamic* models consider a given system through several temporal stages, corresponding to a sequence of decisions. In many instances the temporal dimension is subdivided into discrete intervals of a previously fixed span: minutes, hours, days, weeks, months and years are examples of discrete subdivisions of the time axis.

Development of a model

The development of a mathematical model for decision making into four primary phases



Phases in the development of mathematical models for decision

Problem identification

First of all, the problem at hand must be correctly identified. The observed critical symptoms must be analyzed and interpreted in order to formulate hypotheses for investigation.

Model formulation

Once the problem to be analyzed has been properly identified, effort should be directed toward defining an appropriate mathematical model to represent the system. A number of factors affect and influence the choice of model, such as the *time horizon*, the *decision variables*, the *evaluation criteria*, the *numerical parameters* and the *mathematical relationships*.

Development of algorithms

Once a mathematical model has been defined, one will naturally wish to proceed with its solution to assess decisions and to select the best alternative.

Implementation and test

When a model is fully developed, then it is finally implemented, tested and utilized in the application domain. It is also necessary that the correctness of the data and the numerical parameters entered in the model be preliminarily assessed. These data usually come from a data warehouse or a data mart previously set up. A number of factors should be taken into account at this stage:

- The plausibility and likelihood of the conclusions achieved;
- The consistency of the results at extreme values of the numerical parameters;
- The stability of the results when minor changes in the input parameters are introduced.

Factors affect and influence the choice of model

1. **Time horizon.** Usually a model includes a temporal dimension. For example, to formulate a tactical production plan over the medium term it is necessary to specify the production rate for each week in a year.

2. **Evaluation criteria.** Appropriate measurable performance indicators should be defined in order to establish a criterion for the evaluation and comparison of the alternative decisions. These indicators include the following factors:
   - monetary costs and payoffs;
   - effectiveness and level of service;
   - quality of products and services;
   - flexibility of the operating conditions;
   - reliability in achieving the objectives.

3. **Decision variables.** Symbolic variables representing alternative decisions should then be defined. For example, if a problem consists of the formulation of a tactical production plan over the medium term, decision variables should express production volumes for each product.

4. **Numerical parameters.** It is also necessary to accurately identify and estimate all numerical parameters required by the model. In the production planning example, the available capacity should be known in advance for each process.

5. **Mathematical relationships.** The final step in the formulation of a model is the identification of mathematical relationships among the decision variables, the numerical parameters and the performance indicators defined during the previous phases. Sometimes these relationships may be exclusively deterministic, while in other instances it is necessary to introduce probabilistic relationships.

Classes of models

There are several classes of mathematical models for decision making, which in turn can be solved by a number of alternative solution techniques. Each model class is better suited to represent certain types of decision-making processes.

- **Predictive models**

  A significant proportion of the models used in business intelligence systems, such as optimization models, require input data concerned with future events. As a consequence, *predictive* models play a primary role in business intelligence systems, since they are logically placed upstream with respect to other mathematical models and, more generally, to the whole decision-making process.

  Predictions allow input information to be fed into different decision-making processes, arising in strategy, research and development, administration and control, marketing, production and logistics. Basically, all departmental functions of

an enterprise make some use of predictive information to develop decision making, even though they pursue different objectives.

Predictive models can be subdivided into two main categories. The purpose of *explanatory* models is to functionally identify a possible relationship between a dependent variable and a set of independent attributes. The purpose of *time series* models, is to functionally identify any temporal pattern expressed by a time series of observations referred to the same numerical variable.

- **Pattern recognition and machine learning models**

  the purpose of pattern recognition and learning theory is to understand the mechanisms that regulate the development of intelligence, understood as the ability to extract knowledge from past experience in order to apply it in the future. Mathematical models for learning can be used to develop efficient algorithms that can perform such task.

  Besides an intrinsic theoretical interest, mathematical methods for learning are applied in several domains, such as recognition of images, sounds and texts; biogenetic and medical diagnosis; relational marketing, for segmenting and profiling customers; manufacturing process control; identification of anomalies and fraud detection.

  Mathematical models for learning have two primary objectives. The purpose of *interpretation* models is to identify regular patterns in the data and to express them through easily understandable rules and criteria. *Prediction* models help to forecast the value that a given random variable will assume in the future, based on the values of some variables associated with the entities of a database, as for explanatory models. Based on the existence or not of a *target* attribute, the learning process may be *supervised* or *unsupervised*.

- **Optimization models**

  Many decision-making processes faced by companies or complex organizations can be cast according to the following framework: given the problem at hand, the decision maker defines a set of *feasible* decisions and establishes a criterion for the evaluation and comparison of alternative choices, such as monetary costs or payoffs. At this point, the decision maker must identify the *optimal* decision according to the evaluation criterion defined, that is, the choice corresponding to the minimum cost or to the maximum payoff.

  In general, optimization models arise naturally in decision-making processes where a set of limited resources must be allocated in the most effective way to different entities. These resources may be personnel, production processes, raw materials, components or financial factors. Among the main application domains requiring an optimal allocation of the resources we find:

    - logistics and production planning;
    - financial planning;
    - work shift planning;

- marketing campaign planning;
- price determination.

Mathematical optimization models represent a fairly substantial class of optimization problems that are derived when the objective of the decision making process is a function of the decision variables, and the criteria describing feasible decisions can be expressed by a set of mathematical equalities and inequalities in the decision variables.

- **Project management models**

  A *project* is a complex set of interrelated activities carried out in pursuit of a specific goal, which may represent an industrial plant, a building, an information system, a new product or a new organizational structure, depending on the different application domains. The execution of the project requires a planning and control process for the interdependent activities as well as the human, technical and financial resources necessary to achieve the final goal.

  Project management methods are based on the contributions of various disciplines, such as business organization, behavioral psychology and operations research. Mathematical models for decision making play an important role in project management methods.

  Finally, different classes of optimization models allow the analysis to be extended to the complex problem of optimally allocating a set of limited resources among the project activities in view of execution costs and times.

- **Risk analysis models**

  Some decision problems can be described according to the following conceptual paradigm: the decision maker is required to choose among a number of available alternatives, having uncertain information regarding the effects that these options may have in the future.

  However, the decision maker is forced to make a choice *before* knowing with absolute certainty the level of future demand. At best, she may obtain some stochastic information regarding the likelihood of occurrence of future events by carrying out some market research.

  In situations of this type, the methodological support offered by risk analysis models, primarily based on Bayesian and utility theories, may prove quite helpful. Indeed, this class of models is used successfully in several application domains, such as technology investment, design of new products, research and development, and financial and real estate investment.

- **Waiting line models**

  The purpose of *waiting line* theory is to investigate congestion phenomena occurring when the demand for and provision of a service are stochastic in nature. If the arrival times of the customers and the duration of the service are not known beforehand in a deterministic way, conflicts may arise between customers in the use

of limited shared resources. As a consequence, some customers are forced to wait in a line.

Schematically, a waiting line system is made up of three main components: a *source* generating a stochastic process in which entities, also referred to as customers, arrive at a given location to obtain a service; a set of *resources* providing the service; a *waiting area* able to receive the entities whose requests cannot immediately be satisfied.

Waiting line models allow the performance of a system to be evaluated once its structure has been defined, and therefore are mostly useful within the system design phase.

The main components of a waiting line system are the *population*, the *arrivals process*, the *service process*, the *number of stations*, and the *waiting line rules*. The population, which can be finite or infinite, represents the source from which potential customers are drawn and to which they return once the requested service has been received. The arrivals process describes how customers arrive at the system entry point. In general this is a stochastic process described by the probability distribution of the inter-arrival times, that is, the time intervals between the arrival of two consecutive customers. The service process describes how the providers meet the requests of the customers waiting in line. This in turn is a stochastic process, defined by the probability distribution of the service time, that is, the amount of time that customers spend with the resources providing the service. The number of existing stations and the number of providers assigned to each station are additional relevant parameters of the waiting line system. The waiting rules describe the order in which customers are extracted from the line to be admitted to the service. A primary role is finally played by priority schemes in which a level of priority is assigned to each customer. The customer with the highest priority is then served before all the other customers waiting in line.

Data mining

Definition of data mining

Data mining activities constitute an iterative process aimed at the analysis of large databases, with the purpose of extracting information and knowledge that may prove accurate and potentially useful for knowledge workers engaged in decision making and problem solving.

The data mining process is based on inductive learning methods, whose main purpose is to derive general rules starting from a set of available examples, consisting of past observations recorded in one or more databases.

The purpose of a data mining analysis is to draw some conclusions starting from a sample of past observations and to generalize these conclusions with reference to the entire population, in such a way that they are as accurate as possible.

Data mining activities can be subdivided into two major investigation streams, according to the main purpose of the analysis: *interpretation* and *prediction*.

Interpretation. The purpose of interpretation is to identify regular patterns in the data and to express them through rules and criteria that can be easily understood by experts in the application domain. The rules generated must be original and non-trivial in order to actually increase the level of knowledge and understanding of the system of interest.

Prediction. The purpose of prediction is to anticipate the value that a random variable will assume in the future or to estimate the likelihood of future events. Actually, most data mining techniques derive their predictions from the value of a set of variables associated with the entities in a database.

Differences between OLAP, statistics and data mining

OLAP	statistics	data mining
extraction of details and aggregate totals from data	verification of hypotheses formulated by analysts	identification of patterns and recurrences in data
information	validation	knowledge
distribution of incomes of home loan applicants	analysis of variance of incomes of home loan applicants	characterization of home loan applicants and prediction of future applicants

Applications of data mining

Relational marketing. Data mining applications in the field of relational marketing, have significantly contributed to the increase in the popularity of these methodologies. Some relevant applications within relational marketing are:

- identification of customer segments that are most likely to respond to targeted marketing campaigns, such as *cross-selling* and *up-selling*;
- identification of target customer segments for retention campaigns;
- prediction of the rate of positive responses to marketing campaigns;
- interpretation and understanding of the buying behavior of the customers;
- analysis of the products jointly purchased by customers, known as *market basket analysis*.

Fraud detection. Fraud detection is another relevant field of application of data mining. Fraud may affect different industries such as telephony, insurance (false claims) and banking (illegal use of credit cards and bank checks; illegal monetary transactions).

Risk evaluation. The purpose of risk analysis is to estimate the risk connected with future decisions, which often assume a dichotomous form. For example, using the past observations available, a bank may develop a predictive model to establish if it is appropriate to grant a monetary loan or a home loan, based on the characteristics of the applicant.

Text mining. Data mining can be applied to different kinds of texts, which represent unstructured data, in order to classify articles, books, documents, emails and web pages. Examples are web search engines or the automatic classification of press releases for storing purposes. Other text mining applications include the generation of filters for email messages and newsgroups.

Image recognition. The treatment and classification of digital images, both static and dynamic, is an exciting subject both for its theoretical interest and the great number of applications it offers. It is useful to recognize written characters, compare and identify human faces, apply correction filters to photographic equipment and detect suspicious behaviors through surveillance video cameras.

Web mining. Web mining applications, are intended for the analysis of so-called *clickstreams* – the sequences of pages visited and the choices made by a web surfer. They may prove useful for the analysis of e-commerce sites, in offering flexible and customized pages to surfers, in caching the most popular pages or in evaluating the effectiveness of an e-learning training course.

Medical diagnosis. Learning models are an invaluable tool within the medical field for the early detection of diseases using clinical test results. Image analysis for diagnostic purpose is another field of investigation that is currently burgeoning.

Representation of input data

In most cases, the input to a data mining analysis takes the form of a two dimensional table, called a *dataset*, irrespective of the actual logic and material representation adopted to store the information in files, databases, warehouses and data marts used as data sources. The rows in the dataset correspond to the *observations* recorded in the past and are also called *examples*, *cases*, *instances* or *records*. The columns represent the information available for each observation and are termed *attributes*, *variables*, *characteristics* or *features*.

The attributes contained in a dataset can be categorized as *categorical* or *numerical*, depending on the type of values they take on.

- **Categorical.** Categorical attributes assume a finite number of distinct values, in most cases limited to less than a hundred, representing a qualitative property of an entity to which they refer.
- **Numerical.** Numerical attributes assume a finite or infinite number of values and lend themselves to subtraction or division operations.

Sometimes a more refined taxonomy of attributes can prove useful.

- **Counts.** Counts are categorical attributes in relation to which a specific property can be true or false. These attributes can therefore be represented using Boolean variables {true, false} or binary variables {0,1}.
- **Nominal.** Nominal attributes are categorical attributes without a natural ordering, such as the province of residence.
- **Ordinal.** Ordinal attributes, such as education level, are categorical attributes that lend themselves to a natural ordering but for which it makes no sense to calculate differences or ratios between the values.
- **Discrete.** Discrete attributes are numerical attributes that assume a finite number or a countable infinity of values.
- **Continuous.** Continuous attributes are numerical attributes that assume an uncountable infinity of values.

Data mining process

The main phases of a generic data mining process.

- **Definition of objectives.** Data mining analyses are carried out in specific application domains and are intended to provide decision makers with useful knowledge.
  As a consequence, intuition and competence are required by the domain experts in order to formulate plausible and well-defined investigation objectives. If the problem at hand is not adequately identified and circumscribed one may run the risk of thwarting any future effort made during data mining activities. The definition of the goals will benefit from close cooperation between experts in the field of application and data mining analysts.
- **Data gathering and integration.** Once the objectives of the investigation have been identified, the gathering of data begins. Data may come from different sources and therefore may require integration. Data sources may be internal, external or a combination of the two. The integration of distinct data sources may be suggested by the need to enrich the data with new descriptive dimensions, such as geomarketing

variables, or with lists of names of potential customers, termed *prospects*, not yet existing in the company information system.

- **Exploratory analysis.** In the third phase of the data mining process, a preliminary analysis of the data is carried out with the purpose of getting acquainted with the available information and carrying out *data cleansing*. Usually, the data stored in a data warehouse are processed at loading time in such a way as to remove any *syntactical* inconsistencies.
- **Attribute Selection.** In the subsequent phase, the relevance of the different attributes is evaluated in relation to the goals of the analysis. Attributes that prove to be of little use are removed, in order to cleanse irrelevant information from the dataset. Furthermore, new attributes obtained from the original variables through appropriate transformations are included into the dataset.
- **Model development and validation.** Once a high quality dataset has been assembled and possibly enriched with newly defined attributes, pattern recognition and predictive models can be developed. Usually the *training* of the models is carried out using a sample of records extracted from the original dataset. Then, the predictive accuracy of each model generated can be assessed using the rest of the data. More precisely, the available dataset is split into two subsets. The first constitutes the *training set* and is used to identify a specific learning model within the selected class of models. The second subset is the *test set* and is used to assess the accuracy of the alternative models generated during the training phase, in order to identify the best model for actual future predictions.
- **Prediction and interpretation.** Upon conclusion of the data mining process, the model selected among those generated during the development phase should be implemented and used to achieve the goals that were originally identified. Moreover, it should be incorporated into the procedures supporting decision-making processes so that knowledge workers may be able to use it to draw predictions and acquire a more in-depth knowledge of the phenomenon of interest.

Analysis methodologies

seven basic data mining tasks can be identified:

- **Characterization and discrimination.** Where a categorical target attribute exists, before starting to develop a classification model, it is often useful to carry out an exploratory analysis whose purpose is twofold. On the one hand, the aim is to achieve a characterization by comparing the distribution of the values of the attributes for the records belonging to the same class. On the other hand, the purpose is to detect a difference, through a comparison between the distribution of the values of the attributes for the records of a given class and the records of a different class, or between the records of a given class and all remaining records.
- **Classification.** In a classification problem a set of observations is available, usually represented by the records of a dataset, whose target class is known. Observations may correspond, for instance, to mobile phone customers and the binary class may indicate whether a given customer is still active or has churned. Each observation is described by a given number of attributes whose value is known.

- **Regression.** Unlike classification, which is intended for discrete targets, regression is used when the target variable takes on continuous values. Based on the available explanatory attributes, the goal is to predict the value of the target variable for each observation.
- **Time series.** Sometimes the target attribute evolves over time and is therefore associated with adjacent periods on the time axis. In this case, the sequence of values of the target variable is said to represent a *time series*.
- **Association rules.** Association rules, also known as *affinity groupings*, are used to identify interesting and recurring associations between groups of records of a dataset.
- **Clustering.** The term *cluster* refers to a homogeneous subgroup existing within a population. Clustering techniques are therefore aimed at segmenting a heterogeneous population into a given number of subgroups composed of observations that share similar characteristics; observations included in different clusters have distinctive features. Unlike classification, in clustering there are no predefined classes or reference examples indicating the target class, so that the objects are grouped together based on their mutual homogeneity.
- **Description and visualization.** The purpose of a data mining process is sometimes to provide a simple and concise representation of the information stored in a large dataset.

Data preparation

Data validation

The quality of input data may prove unsatisfactory due to *incompleteness*, *noise* and *inconsistency*.

> **Incompleteness.** Some records may contain missing values corresponding to one or more attributes, and there may be a variety of reasons for this. It may be that some data were not recorded at the source in a systematic way, or that they were not available when the transactions associated with a record took place. It is also possible that some data were deliberately removed during previous stages of the gathering process because they were deemed incorrect. Incompleteness may also derive from a failure to transfer data from the operational databases to a data mart used for a specific business intelligence analysis.
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> **Noise.** Data may contain erroneous or anomalous values, which are usually referred to as *outliers*. Other possible causes of noise are to be sought in malfunctioning devices for data measurement, recording and transmission. The presence of data expressed in heterogeneous measurement units, which therefore require conversion, may in turn cause anomalies and inaccuracies.
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> **Inconsistency.** Sometimes data contain discrepancies due to changes in the coding system used for their representation, and therefore may appear inconsistent.

The purpose of data validation techniques is to identify and implement corrective actions in case of incomplete and inconsistent data or data affected by noise.

Incomplete data

To partially correct incomplete data one may adopt several techniques.

- **Elimination.** It is possible to discard all records for which the values of one or more attributes are missing. In the case of a supervised data mining analysis, it is essential to eliminate a record if the value of the target attribute is missing. A policy based on systematic elimination of records may be ineffective when the distribution of missing values varies in an irregular way across the different attributes, since one may run the risk of incurring a substantial loss of information.
- **Inspection.** Alternatively, one may opt for an inspection of each missing value, carried out by experts in the application domain, in order to obtain recommendations on possible substitute values. Obviously, this approach suffers from a high degree of arbitrariness and subjectivity, and is rather burdensome and time-consuming for large datasets.
- **Identification.** As a third possibility, a conventional value might be used to encode and identify missing values, making it unnecessary to remove entire records from the given dataset.
- **Substitution.** Several criteria exist for the automatic replacement of missing data, although most of them appear somehow arbitrary. For instance, missing values of an attribute may be replaced with the mean of the attribute calculated for the

remaining observations. This technique can only be applied to numerical attributes, but it will clearly be ineffective in the case of an asymmetric distribution of values. In a supervised analysis it is also possible to replace missing values by calculating the mean of the attribute only for those records having the same target class.

Data transformation